MILITARY SPECIFICATION

LIFE PRESERVER, UTILITY, AUTO-INFLATABLE, MK 5, MOD O

This specification is approved for use by the Naval Sea Systems command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for an Automatically Inflatable Utility Life Preserver (AIULP), MK 5, MOD O, referred to as preserver hereafter, which is to be worn by personnel during utility type work situations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1894	-	Paper, Kraft, Treated (Fire Resistant)
L-P-375	-	Plastic Film, Flexible, Vinyl Chloride
V-F-106	-	Fastener, Slide, Interlocking
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage, (General Use)
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: 05Q42, 2531 National Center Bldg. 3, Washington, DC 20362-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this specification or by letter.

AMSC N/A FSC 4220
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MIL-P-116	- Preservation, Methods of
MIL-T-5038	Tape Textile and Webbing, Textile, Reinforcing, Nylon
MIL-C-5040	- Cord, Fibrous, Nylon
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MS20230	- Grommets, Metallic, Plain & Spur, with Washers, Type I and III
MIL-F-21840	- Fastener Tapes, Hook and Loop, Synthetic
MIL-C-25369	- Cylinders, Carbon Dioxide, for Inflation of Pneumatic Life Preservers
MIL-C-43989	- Cloth, Knitted, Raschel, Aramid
MIL-O-81375	- Oral Inflation Assemblies, Survival Equipment, Inflatable
MIL-T-83193	- Thread, Aramid, Spun Staple
MIL-C-83429	- Cloth, Plain and Basket Weave, Aramid

STANDARDS

FEDERAL

FED-STD-595	Colors Used in Government Procurement
FED-STD-751	- Stitches, Seams, and Stitching

MILITARY

MIL-STD-2073-1	- DoD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards , and handbooks are available from the Standardization) Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings. and publications . The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

u. s. COAST GUARD

Code of Federal Regulations, 46 Part 164.018
Code of Federal Regulations, 46 Part 33.38-5

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-2482705	- Life Preserver, Vest, MK 1, Main Assembly

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

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NAVAL AIR SYSTEMS COMMAND (NAVAIR)
68A94C13 - Light Assembly, Survivor Locator

(Application for copies should be addressed to: Commander, Naval Air Development Center, Code 603132, Warminster, PA 18974-5000.)

2.2 Non-Government publications . The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
AMS 3272 - Cloth, Nylon, Polyurethane Coated Thermally, Radio Frequency, or Ultrasonically Sealable
AMS 3272/1 - Cloth, Nylon, Polyurethane Coated Plain Weave, 6.0 oz per sq yd (200 g/m^2)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc. , 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence, Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample or samples of the AIULP MK 5 MOD O shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 General description. The AIULP M.K 5, MOD O, shown in figure 1, is an automatic and manual $C0_2$ inflatable life preserver. The bladder is enclosed in a fire retardant vest constructed of aramid and heavy nylon cloth. A single, broad, nylon webbing is provided to secure the vest to the body. A non-metallic zipper and buckle are provided for closure of the preserver front to ensure the inflated bladder remains in the desired position in the upper chest area. The size adjustment webbing allows for fit of 811 sizes. The vest closure shall be hook and pile, and shall not restrict the bladder from emerging from the cover when

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inflated. A hood with face shield is also provided to protect the user from splashing waves.



FIGURE 1. Auto Inflatable Utility Life Preserver MARK 5. MOD 0.

3.3 Materials and components. The materials and components shall conform to applicable specifications, standards, and patterns (see 3.5) required herein. Unless otherwise specified, the non-metallic materials used in construction of the inflation assembly shall not have been manufactured more than 24 months prior to the date of delivery except for rubber products which shall have a cure date more than 12 months prior to the delivery date. The components are the aramid vest and the inflatable bladder assembly.

3.3.1 Vest. The vest shall be constructed of the following materials and all components shall be Green, color 34088 of FED-STD-595 unless otherwise stated.

(a) Aramid mesh. The back and sides shall be constructed of aramid raschel knit mesh conforming to MIL-C-43989.
(b) Basic material. The base fabric shall be plain weave, aramid cloth conforming to Type II, class 1 of MIL-C-83429.
(c) Tape . woven binding. The cloth binding for the mesh, casing, hood, and pocket covers shall be constructed of MIL-T-5038, Type III.
(d) Fastener tapes . The hook and pile fastener tape shall conform to MIL-F-21840, Type II, class 1. The fastener tape shall be attached in locations as shown on the patterns.
(e) Straps and side adjustment webbing. Straps and webbing shall be constructed of nylon conforming to MIL-T-5038, Type IV.

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(f) Anti-splash face shield. The face shield shall be constructed of 0.020-inch transparent vinyl plastic conforming to Type II, class 1 of L-P-375.

(g) Grommets. Brass grommets used in the vest and bladder shall conform to MS20230, B10 and B20.

(h) Slide fastener. The slide fastener shall be molded plastic, individual element or continuous element, monofilament, conforming to Type IV Style 8 Size MHS, of V-F-106. The tape shall be synthetic or cotton/synthetic in the color black or green, approximating the shade of the basic fabric.

(i) Side adjustment hardware. The side adjustment strap shall be foward pull containing a one half inch adjustment buckle, Waterbury part no. 104-0150 or equal, manufactured from virgin acetal resin with not more than 20 percent (%) of reground of the same material. The fastener and its adjusting device shall withstand a 175 pound (lb) pulling force for 15 seconds using the appropriate webbing.

(j) Buckle . The front closure buckle shall be a one and a half inch, side release buckle, Waterbury part no. 101-0150 or equal, manufactured from virgin acetal resin with not more than 20% of reground of the same material. The fastener shall withstand a 175 lb pulling force for 15 seconds using the appropriate webbing.

3.3.2 Inflatable bladder.

(a) Coated cloth. The coated cloth used in construction of the inflatable bladder and blast patch shall be heat-sealable polyurethane coated nylon conforming to SAE AMS 3272 and AMS 3272/1 except that tear strengths shall be: 3.75 pounds for warp, and 2.75 pounds for fill . The coated cloth color shall be Orange 32246 of FED-STD-595. The blast patch shall be circular in shape with a diameter of 3 inches ± 1/8 inch.

(b) Manifold. The inflation device shall be securely attached to the preserver by means of a brass manifold with a heat sealable base, which shall in turn be heat sealed to the inside surface of the bladder fabric, so that the inflation assembly projects on the outside. The manifold shall conform to NAVSEA Drawing 803-2482705.

(c) Bladder laces. The cord used to lace the bladder to the casing shall be nylon MIL-C-5040, Type III.

(d) Retro-reflective tape. Reflective tape for the front casing, hood and bladder shall conform to US Coast Guard, “Code of Federal Regulations” 46 Part 164.018 Type I.

(e) Oral inflation valve and tube. The oral inflation valve and tube shall be Type II, class 1, style E of MIL-O-81375 and shall be 4 1/8 inches ± 1/8 inch.

(f) Adhesive. The adhesive for securing the parts to the bladder shall be chloroprene and conform to 3M PN 2141PT or equal.

3.4 Accessories. The AIULP MK 5, MOD O shall have the following accessories :

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(a) Whistle. Each vest shall be equipped with a nonmetallic, USCC approved, whistle secured to the vest by a 22 inch long lanyard cord. The whistle shall be of a flat design and produce a loud shrill dual tone. The whistle shall conform to "Code of Federal Regulations" 46 Part 33.38-5. The dimensions shall be no greater than 2 inches long by 7/8 inch wide and 5/16 inch deep at the air chamber .

(b) Auto lamp unit. Each vest shall be equipped with an auto lamp unit which is powered by a water activated battery conforming to NAVAIR drawing number 68A94C13, with the following exception: length of the lead wire shall be 20 inches ± 1 inch.

(c) Buddy life line. Each vest shall be equipped with a l-inch wide, 66 inch long (finished length) buddy life line attached according to mark on left pocket flap pattern. The line shall conform to MIL-T-5038, Type IV.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be within ± 1/16 inches of the Government patterns. Neither the Government patterns, nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns.

3.5.1 Pattern Parts. The component parts shall be cut from materials as specified, and in Table I. No drill holes are allowed.

TABLE I. Patterns.

Pattern Parts	Material	Cut Parts	Markers
Pocket Divider	Green Woven Material (MIL-C-83429)	2	
Flap - Right Side	Green Woven Material (MIL-C-83429)	1	
Flap - Left Side	Green Woven Material (MIL-C-83429)	1	
Pocket	Green Woven Material (MIL-C-83429)	2	
Bladder Casing Front	Green Woven Material (MIL-C-83429)	1	
Bladder Casing Back	Green Woven Material (MIL-C-83429)	1	
Hood Casing	Green Woven Material (MIL-C-83429)	2	
Pocket Tunnel Piece	Green Woven Material (MIL-C-83429)	2	

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TABLE X. Patterns (cent'd) .

Pattern Parts	Material	Cut Parts	Markers
Bladder Cylinder Holder	Orange Coated Cloth (SAE AMS 3272 & 3272/1)	1	
Bladder Lamp Piece	Orange Coated Cloth (SAE AMS 3272 & 3272/1)	1	
Bladder	Orange Coated Cloth (SAE AMS 3272 & 3272/1)	2	
Back	Raschel Knit Material (MIL-C-43989)	1	
Front	Raschel Knit Material (MIL-C-43989)	2	
Transparent Plastic Shield	Vinyl Plastic (L-P-375 Type II, Class 1)	1	
Bladder Marker - Front View			1
Bladder Marker - Back View			1
Raschel Knit Vest Marker			1
Hood Casing Marker			1
Bladder Casing Front Marker			1

3.6 Recovered materials. Unless otherwise specified herein, all equipment, material , and articles incorporated in the products covered by this document shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this document unless otherwise specifically specified.

3.7 Construction. The preserver shall be constructed in accordance with the operations shown in Table II. The contractor is not required to follow the exact sequence of operations provided that the finished vests are identical to those produced by following the sequence as listed in Table II.

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TABLE II. Construction.

No.	Operation	Stch type	Seam/ S tch type	Stch\ in	Thread ndl bob/lpr
1.	Make Bladder.				
	(a) Heat seal the inflation manifold device to the back side of the front ply of the bladder so that the infla- tion assembly projects out through the face according to the mark on the blad- der marker - front view.				
	(b) Apply an identification label to the back right side of the inflation bladder - placement as shown on bladder marker-back view.				
	(c) Apply an inspection record label to the back left side of the inflation bladder - placement as shown on bladder marker-back view.				
	(d) Apply reflective tape to five loca- tions on front ply of bladder, using specified adhesive, according to marks on bladder marker - front view.				
	(e) Apply pile portion of fastener to four locations on front ply of bladder, using suitable adhesive, according to marks on bladder marker - front view.				
	(f) Heat seal the oral inflation tube to right side of front according to mark on bladder marker - front view.				
	(g) Apply holder for $C0_2$ cylinder On back ply of bladder using a suitable adhesive, according to mark on bladder marker - front view.				
	(h) Heat seal the blast patch to the back side of the back ply of bladder, using suitable adhesive according to marks on bladder marker - back view.				

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TABLE II. construction (cont'd)

No.	Operation	Stch type	Seam/ stch type	Stch/ in	Thread ndl bob/lpr
	(i) Heat seal the two bladder plies together, back to back 1/2 inch from edge. The distance from cut edge of bladder to the inner-most edge of heat seal shall be wider at areas where extensions protrude from the bladder.				
	(j) Apply ten grommets to the inner side and neck extensions according to marks on bladder marker - front view.				
	(k) Fold automatic lamp unit holder in half forming a loop, and inserting between seam allowance, stitch to outer extension on left side of bladder, with two rows of stitching, according to marks on bladder marker - front view. Backstitch ends of stitching.	301	SSa-1	8-12	
2.	Make Front Pockets.				
	(a) Stitch bottom corners of front pocket, face to face.	301	SSa-1	8-12	
	(b) Overedge all remaining edges of front pockets.	503 or 504 or 602	EFd-1	6-10	
	(c) Form front pocket hem, turning edge under 5/8 inch, and stitch 1/4 inch from folded edge.	301	EFa-1	8-12	
	(d) Stitch sides of front pocket, back to back pocket patterns, forming a welt, from hem to the end of seam stitched in operation 2(a).	301	Sf-1	8-12	
	(e) Position pile portion of fastener to tape on front pocket according to mark on pattern, and stitch all four sides.	301	Sbj-1	8-12	
	(f) Bind all sides of pocket divider, overlapping binding a minimum of 3/4 inch.	301	Sc-2	8-12	

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TABLE II. Construction (cent'd).

No.	Operation	Stch type	Seam/ stch type	Stch/ in	Thread ndl bob/lpr
	(g) Stitch one edge of pocket divider to pocket according to mark on pocket pattern.	301	Sbd-1	8-12	
	(h) Attach a grommet to each side of pocket, according to marks on pocket pattern.				
3.	Make Front Pocket Flaps.				
	(a) Fold pocket flap in half back to back, and bind all raw edges.	301	Se-l	8-12	
	(b) Make a pull tape for pocket flap by folding a 3 inch long piece of 3/4 inch webbing in half, and edge stitch all sides.	301	Sa-1 and Osf-1	8-12	
	(c) Position pull tab on back of hook tape, with free end extending 3/4 inch beyond box-stitch edge of tape, accord-ing to mark on pocket flap pattern. Attach with a 1/2 inch cross boxstitch.	301	cross-boxsti tch	8-12	
	(d) Position hook tape on inside of flap according to mark on pocket pat-tern, with pull tab, extending beyond flap.	301	Sbj -1	8-12	
4.	Make Vest.				
	(a) Apply an identification label to the right side of the front bladder casing (underside) - placement as shown on bladder casing front pattern.				
	(b) Join back to front at shoulder with a double lapped, double stitched seam, 1/2 inch gauge.	301	SC-2	8-12	
	(c) Join fronts to back at side seams, with double lapped, double stitched seam, 1/2 inch gauge.	301	SC-2	8-12	

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TABLE II. Construction (cont'd).

No.	Operation	Stch type	Seam/ stch type	Stch/ in	Thread ndl bob/lpr
	(d) Position piece of 3/4 inch woven webbing over the seams constructed in operations 4(b) and (c). Stitch 1/4 inch from each edge of webbing through all plies.	301	similar to SSad-2	8-12	
	(e) Stitch reflective tape to four locations to outside of the front bladder casing according to marks on front bladder casing marker.	301	LSbj-1	8-12	
	(f) Bind armholes, overlapping binding minimum of 3/4 inch with raw edges turned under. Bind edge of vest starting and finishing at mark on Raschel knit front pattern.	301	BSc-1	8-12	
	(g) Bind opening in center of back, overlapping binding a minimum of 3/4 inch, with the raw edges turned under.	301	BSc-1	8-12	
	(h) Insert each end of the 10 inch long piece of 1-1/2 inch webbing through a slide adjustment hardware. Fold the webbing ends back under itself approximately 1-1/2 inches, and secure each end with a 3/4 by 1 inch cross stitch. Center on vest back according to marks on Raschel knit back. Edgestitch along top and bottom and stitch 3/4 inch from webbing fold. The distance between the outer edges of the two boxstitches shall be approximately 4-1/4 inches.	301	LSbj-2	8-12	
	(i) Overedge all sides of the pocket tunnel pieces.	503 or 504 or 602	EFd-1	6-10	
	(j) Turn side edges of tunnel piece back 3/8 inch, face to face, and edgestitch.	301	EFa-1	8-12	

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TABLE II. Construction (cent'dl.

No.	Operation	Stch type	Seam/ stch type	Stch/ in	Thread ndl bob/lpr
	(k) Turn under side edges of pocket 3/8 inch. Align folded side edges of tunnel (op. 4 (j)), with folded edges of pocket, and edgestitch along width of tunnel, from notch to 1/4 inch from top edge of tunnel. The top of the tunnel shall be positioned approximately 1-1/4 inches down from top of pocket hem.	301	SSa-1	8-12	
	(1) Fold tunnel piece in half at notch. Position pocket onto vest, according to marks on Raschel knit front, and edgestitch the sides of the bottom ply of the tunnel to the vest from the notch to the top of the tunnel piece. Do not catch the top ply of tunnel or pocket in stitching.	301	LSbj -1	8-12	
	(m) Place top ply of tunnel piece in finished position, aligning top edges, and stitch 1/4 inch across top edge, through both pieces of tunnel and vest, simultaneously catching the straps made of 1 inch webbing under the stitching. Two straps shall be positioned towards the front of the pocket, and one positioned towards the back of the pocket. DO not catch the pocket piece in the stitching The straps shall be inserted no less than 1/2 inch under the stitching. The lengths of the straps shall be 12 inches.	301	LSbj -1	8-12	
	(n) Attach pocket divider to vest, according to marks on Raschel knit front, leaving an opening for the tunnel piece. Backstitch each end of stitching.	301	LSbj -1	8-12	

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TABLE II. Construction (cent'd).

No.	Operation	Stch type	Seam/ stch type	Stch/ in	Thread ndl bob/1pr
	(o) Turn under bottom of pocket 3/8 inch. Attach sides and bottom of pocket, leaving the tunnel open on the sides. Backstitch each end of stitching.	301	LSbj -1	8-12	
	(p) Place vertical bartacks at each end of tunnel opening, maintaining opening of approximately 1-3/4 inches.	Bar-tack	EFB-1		
	(q) Attach buckle and strap assembly to front of vest, according to marks on Raschel knit back pattern. The female portion of the buckle shall be on the right front, with the male portion on the left front. The buckle portion extends beyond the vest assembly. The end of a 46 inch long by 1-1/2 inch wide webbing shall be inserted through the buckle, the webbing folded back approximately 1-1/2 inches, and attached to the vest by stitching a 3/4 x 1 inch cross-boxstitch through all plies. Insert webbing through tunnel, smoothing out webbing and vest. At side seam, attach webbing to vest, with a 3/4 x 1 inch cross-boxstitch.	301	LSbj -1	8-12	
	(r) Insert webbing through the buckles (op. 4 h), and return to front of vest through the tunnel. Fold unfinished end of webbing under against itself, and secure with a 3/4 x 1 inch cross-boxstitch.	301	EFb-1	8-12	
	(s) With pocket flap in open position, stitch flap to vest according to marks on Raschel knit front marker, stitching 1/8 inch from folded edge. The right flap has an indentation on the front portion of the flap (closest to front opening of vest).	301	LSbj -1	8-12	

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TABLE II. Construction (cent'd) .

No.	Operation	Stch type	Seam/ stch type	stch/ in	Thread ndl bob/lpr
	(t) Attach a 66 inch long piece of 1 inch webbing to the inside of the left front pocket flap per webbing attachment mark noted on PKT FLAP-LFT. Enclose the raw edge of webbing and secure with a 3/4 inch cross-boxstitch.	301	LSbj -1	8-12	
	(u) At remaining end of strap, form a 5 inch loop, with raw edge turned under 1 inch. Bartack with a 3/4 inch long bartack through all plies.	Bar-tack	BFb-1		
	(v) Attach a 22 inch cord to whistle, and knot at one end. Take the remaining end and knot the whistle assembly to the grommet on right pocket (near indentation) .				
5.	Make hood.				
	(a) Join hood panels together at center seam.	301	LSq-2(a)	8-12	
	(b) Bind center seam.	301	BSC-1	8-12	
	(c) Turn seam allowance to the right, and topstitch 1/8 inch from folded edge .	301	LSq-2(b)	8.12	
	(d)- Bind face opening.	301	BSC-1	8-12	
	(e) Position face shield to inside of hood and stitch, placing the stitching on the binding stitching of operation 5(d).	301	LSbj -1	8-12	
	(f) Bind ear openings.	301	BSC-1	8-12	
	(g) Bind outer raw edge of hood.	301	BSC-1	8-12	
	(h) Stitch hook portion of fastener tape to inside of the hood in two locations according to marks on hood casing marker pattern.	301	LSbj -1	8-12	
	(i) Stitch loop portion of fastener tape to inside of the hood in two locations according to marks on hood casing marker patterns.	301	LSbj -1	8-12	

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TABLE II. Construction (cent'd~.

No	Operation	Stch type	Seam/ stch type	Stch/ in	Thread ndl bob/lpr
	(j) Stitch reflective tape to outside of the hood according to marks on hood casing marker pattern.	301	LSbj -1	8-12	
	(k) Using a 12 inch long piece of 1-inch bartack webbing, make a hood strap by forming a 3 inch loop with raw edge turned under and bartack with a 3/4 inch bartack.	Bar-tack	EFb-1		
	(1) Attach remaining unfinished end of strap to center front over center seam with raw edge turned under. Secure to hood with a 3/4 inch bartack The strap shall finish 8 ± 1/2 inches long.	Bar - tack	EFb-1		
6.	Make bladder casing.				
	(a) Overedge inner raw edges of back and front plies of bladder casing.	503 or 504 or 602	EFd- 1	8-12	
	(b) Bind bottom and outer raw edge of each ply of bladder casing.	301	BSC-1	8-12	
	(c) Attach hook portion of fastener cape to the back of the front ply of the bladder casing on the bottom and outer edge, with the outer edge of the tape even with the outer edge of the binding. Stitch 1/8 inch from outer edge .	301	LSbj -1	8-12	
	(d) Attach loop portion of fastener tape to the back of the back ply of the bladder casing on the bottom and outer edge, with the outer edge of the tape even with the outer edge of the binding. Stitch 1/8 inch from outer edge.	301	LSbj -1	8-12	
	(e) Attach hook portion of fastener tape in two locations to the back of the front ply according to marks on bladder casing marker pattern.	301	LSbj -1	8-12	

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TABLE II. Construction (cent'dl.

No.	Operation	Stch type	Seam/ stch type	stch/ in	Thread ndl bob/lpr
	(f) Form a 5/8 inch hem on mirror pocket.	301	EFa -1	8-12	
	(g) Using elastic webbing, form a loop for light battery and attach to back of back ply of casing according to marks on bladder casing marker pattern. Stitch across the width of the loop.	301	LSbj -1	8-12	
	(h) Attach four grommets to each of two 4-1/2 inch lengths of 1-1/2 inch webbing. Attach 2 grommets to a 2-1/2 inch length of 1-1/2 inch webbing.				
	(i) (Optional) Baste grommet piece in place on each side of back casing, according to marks on bladder casing marker pattern.				
7.	Assemble vest.				
	(a) (Optional) Baste slide fastener to each front according to marks on Raschel knit vest marker pattern.				
	(b) Join front ply of casing, vest, and back ply of casing along inside edge with a 1/2 inch seam, simultaneously catching the slide fastener tapes and grommet pieces.	301	LSak-2(a)	8-12	
	(c) Turn front ply against the seam allowance and edgestitch, 1/8 inch from folded edge.	301	LSak-2(b)	8-12	
	(d) Stitch hood to back of front ply of casing, across the width of the hood, matching marks on Raschel knit vest marker pattern.	301	LSbj -1	8-12	
	(e) Attach back ply of casing to vest according to marks on Raschel knit vest marker pattern.	301	LSbj -1	8-12	

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TABLE II. Consstruction (cent'd\.

No.	Operation	Stch type	Seam/ Stch type	Stch/ in	Thread ndl bob/lpr
	(f) Lace bladder grommets and casing grommets together with cord.				
	(g) Position automatic lamp unit and battery into holder.				

3.7.1 Cut edges. All the cut edges of the uncoated nylon tapes (including the nylon fastener tapes), webbings, and cords shall be seared, prior to the fabrication of the life preserver, to prevent fraying. No sharp edges shall be formed.

3.7.2 Thread. The thread shall be an aramid thread conforming to MIL-T-83193. The color of the thread shall approximately match the color of the basic fabric.

3.7.3 Seams and stitching. No stitching shall pass through any air retaining compartment. All machine stitching in the accessory parts shall be as specified in Table II. All sewing shall conform to FED-STD-751 as specified in Table II. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching in any row shall be maintained within a tolerance of ± 1/16 inch. The ends of the stitching shall be backstitched, by overlapping on itself, a minimum of 1/2 inch. Thread breaks, skips, and run-offs shall be overstitched not less than 1 inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. No seam shall be twisted, puckered, or pleated and no portion of the accessory parts shall be caught in an unrelated operation or seam. All the thread ends shall be trimmed to a length of 1/4 to 3/8 inch. The seam edges shall be properly forced out and shall not contain any folds.

3.7.4 Barracking. The number of stitches per bartack shall be based proportionally on 1/4 inch long bartack containing a minimum of 14 stitches. The length and location of each bartack shall be as specified in Table II.

3.7.5 Heat sealing. A radio frequency (RF) dielectric heat-sealing process shall be used in the construction of the bladder. The distance from the cut edge of the bladder to the inner-most edge of sealed portion shall not exceed 3/4 inch. The width of the heat seal itself shall not be less than 1/8 inch.

3.8 Identification Label. Identification labels shall be provided for both the vest and bladder. The label shall conform to Type IV, Class 11 of DDD-L-20 and shall have the following inscription and size of characters:

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Inscription	Size of characters
Item Identification	18 point Gothic bold
Manufacturer's Name	10 point Gothic medium
Contract Number	10 point Gothic medium
Date of Manufacture (month and year)	10 point Gothic medium
Serial Number	10 point Gothic medium
Lot Number	10 point Gothic medium
National Stock Number	10 point Gothic medium

3.8.1 Item Identification. The item identification for bladder shall be BLADDER, AUTO-INFLATABLE UTILITY LIFE PRESERVER (AIULP) Mark 5, Mod O and the identification for the vest shall be VEST, AUTO-INFLATABLE UTILITY LIFE PRESERVER (AIULP) Mark 5, Mod O.

3.8.2 Serial number. Serialization of the bladder and vest shall be from a block of consecutive numbers covering the entire acquisition quantity.

3.9 Inspection Record Label. The Inspection label shall be 3 1/2 inches wide by 7 1/2 inches long and conform to Type IV, Class 11 of DDD-L-20. Its inscription and size of letters shall be as shown in Figure 2.

INSPECTION RECORD					
TEST	DATE	SIGNATURE	TEST	DATE	SIGNATURE
FUNCTIONAL & LEAK			FUNCTIONAL & LEAK		
LEAK			LEAK		
LEAK			LEAK		
LEAK			LEAK		
FUNCTIONAL & LEAK			FUNCTIONAL & LEAK		
LEAK			LEAK		
LEAK			LEAK		
LEAK			LEAK		
FUNCTIONAL & LEAK			FUNCTIONAL & LEAK		
LEAK			LEAK		
LEAK			LEAK		
LEAK			LEAK		

FIGURE 2. Inspection Record Label.

3.10 Performance characteristics.

3.10.1 Adhesion. Five seam specimens similar to the seams of the bladder shall be subjected to the adhesion test specified in 4.7.2.2. The adhesion, coated surface to coated surface, RF bonding-no adhesive, shall be 45 pounds per inch of width. Specimens do not have to be fully constructed bladders.

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3.10.2 Operation (Carbon Dioxide). When the life preserver is inflated for operations test as specified in 4.7.2.1, design shape shall be attained within 10 seconds. There shall be no leakage of carbon dioxide, no hindrance to the flow of carbon dioxide from the inflation assembly or evidence of material or structural failure in any respect. The seams of attachments shall not pucker when the compartment is inflated or deflated. Upon completion of this test and examination, and while still inflated with carbon dioxide, the preserver shall be subjected to the buoyancy test specified in 4.7.2.3.

3.10.2.1 Oral inflation assembly operation. When inspected as specified in 4.7.2.1, there shall be no hindrance to the flow of carbon dioxide when the valve is depressed and the oral inflation assembly shall provide a positive shut-off when the valve is released.

3.10.3 Buoyancy. When inspected as specified in 4.7.2.3, the life preserver, while inflated with the carbon dioxide, shall support a 32 pound solid steel or lead weight without the entire life preserver assembly sinking below the surface of the water.

3.10.4 Over Pressure. When inspected as specified in 4.7.2.4, the pressure in the bladder shall not be less than 4.5 pound per square inch (psi) . There shall be no evidence of structural or material failure.

3.10.5 Leakage. When inspected as specified in 4.7.2.5, the pressure in the bladder shall not be less than 1.5 psi. There shall be no evidence of structural or material failure.

3.10.6. Visual Examination. When examined as specified in 4.7.1.1, the preservers shall conform to the requirements of this specification and applicable patterns. The classification of defects for visual examination of the life preserver (Table VI) shall be used to classify the defects found,

3.11 Folding of the life preserver. Upon satisfactory completion of all tests and examinations, the life preserver shall be folded in preparation for packaging. The bladder shall be completely evacuated of air and folded in the following manner:

(a) Ensure accessories are properly attached in accordance with patterns (see 3.4).
(b) Ensure nylon cord attaching bladder to vest is securely tied.
(c) Lay preserver flat with pockets facing down and side and top grommets exposed.
(d) Fold hood in half so face shield is flat and pull strap is extended as shown on figure 3.
(e) Roll hood tightly, cover with flap, then fasten with hook-and-pile at neck edge.
(f) Roll top section of bladder, then fasten hook-and-pile to enclose bladder and hood, as shown on figure 4,
(g) Fold both bottom sections of bladder up against bladder as shown on figure 4.

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(h) Fold sides and cop of bladder in accordion type fold as shown on figure 5 (do not roll). Fasten hook-and-pile around sides simultaneously until bladder is completely enclosed.

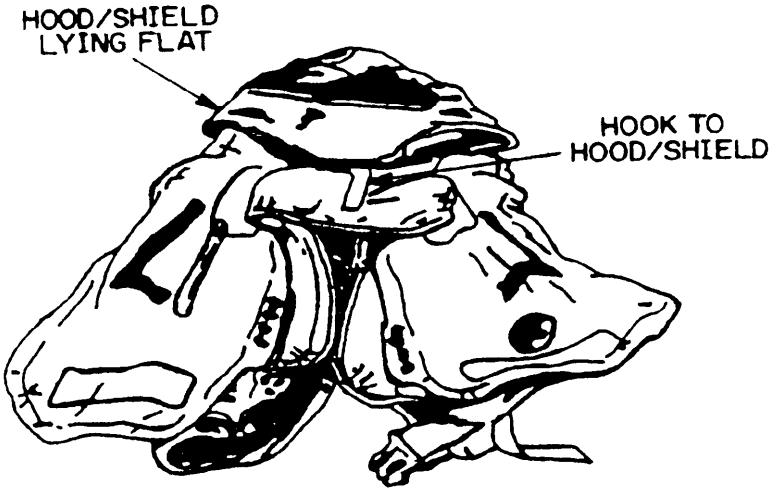


FIGURE 3. Packaging preparation of the preserver.

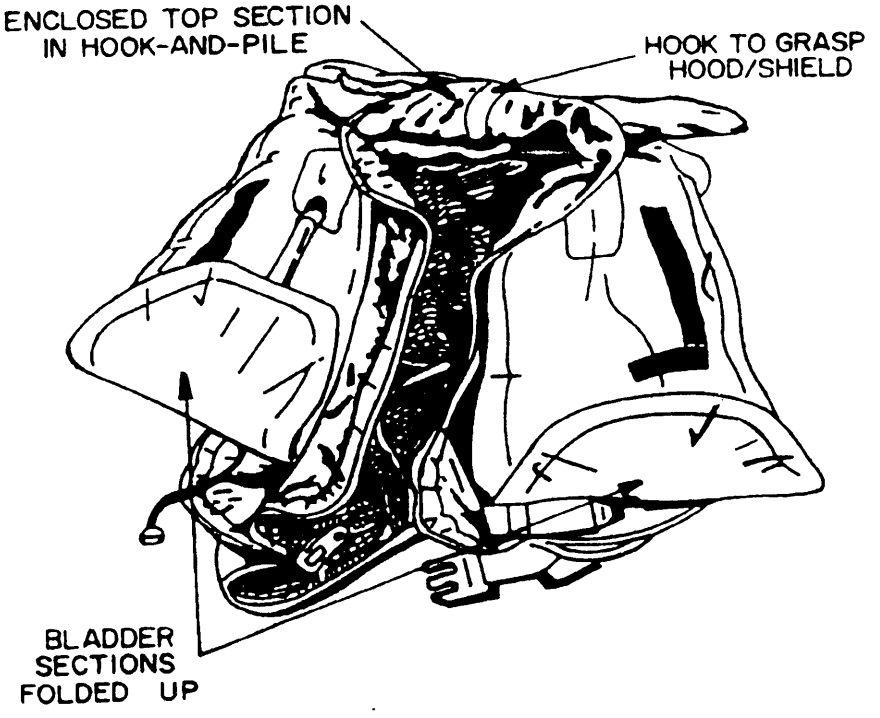


FIGURE 4. Bottom Portion of bladder folded UP.

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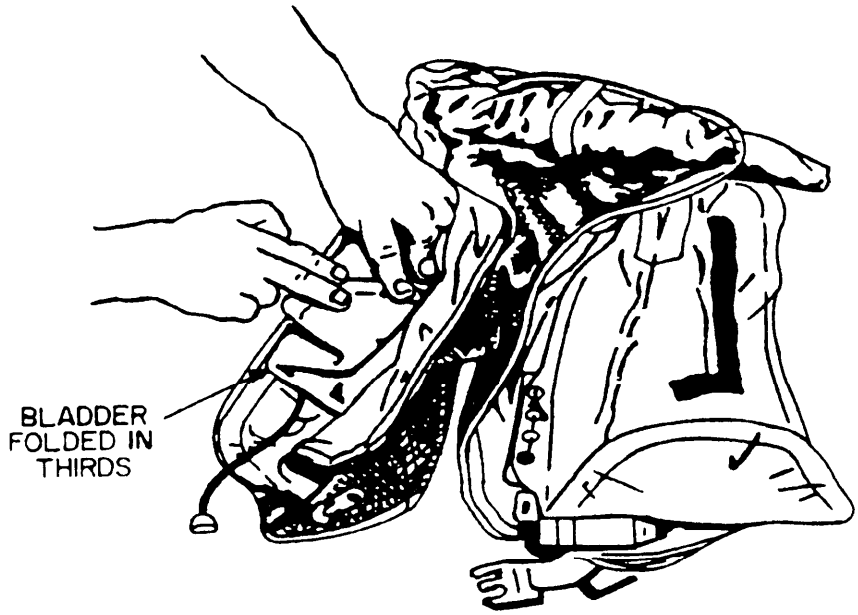


FIGURE 5. Enclosing bladder in cover.

3.12 Workmanship. The life preservers shall be free from all loose thread, lint, and foreign matter. The life preservers shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability or durability. The life preservers shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4),
(c) Quality conformance verification inspection (see 4.5).

4.3 First article inspection. First article inspection shall be conducted at a laboratory specified by the government (see 6.2) . The manufacturer shall submit three complete life preservers and a sample of heat-sealed bladder material as required by 4.5.1 of AMS 3272 (see 6.5). The samples shall be representative of the construction, workmanship, components and materials to be used during production. Approval of first article samples does not exempt the requirements to perform the quality conformance inspections. First article inspection shall consist of the examination and tests specified in Table III. The examinations and tests shall be performed in the sequence listed.

4.3.1 First article inspection samples. Upon completion of the first article inspection program, one approved life preserver will be returned to the manufacturer for use in monitoring production. One life preserver will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract. The other life preserver will be retained by the inspection laboratory for reference and will be returned to the manufacturer with the samples from the final lot as required by 4.5.1.

TABLE III. First article inspection. 1/

Inspection	Test Method
operation with inflation assembly and $C0_2$ cylinder	4.7.2.1
Buoyancy	4.7.2.3
Over Pressure	4,7.2.4
Leakage	4.7.2.5
Visual examination	4.7.1.1

1/ The results of inspection shall be identifiable by the assigned serial numbers .

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the tests specified in Table IV (see 6.4).

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TABLE IV. Quality conformance inspection. 1/

Test Method	Defect	Classification
Visual examination of life preserver	4.7.1.1	Table VI 2/
Operation with inflation assembly and $C0_2$ cylinder	4.7.2.1	Major
Adhesion 5/	4.7.2.2	Minor
Buoyancy 4/	4.7.2.3	Minor
Over Pressure 3/	4.7.2.4	Major 2/
Leakage 3/	4.7.2.5	Major 2/
Packaging	4.7.1.2	Minor

1/ The results of inspection shall be identifiable by the assigned serial numbers .
2/ Test/examine all life preservers in the lot.
3/ These inspections shall be performed in the listed sequence.
4/ The buoyancy test shall be conducted only if the heat seal die is changed between first article test and a lot.
5/ Test shall be five (5) specimens similar to the seams of the bladder. Sampling as specified in table V not required.

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1 .1 Life preservers. An inspection lot size shall be expressed in units of one life preserver made under the same conditions and from the same materials and components. The sample unit shall be one life preserver.

4.4.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container. The sample unit shall be one shipping container fully prepared for delivery with the exception that it not be sealed.

4,4.1.2 Sampling for inspection. As a minimum the contractor shall select a sample quantity from each lot of life preservers and shipping containers in accordance with Table V (except that all vests in the lot shall be tested/examined as noted in Table IV) and inspect them. The sample size for major and minor characteristics is shown in Table V. If one or more major defects are found in any sample, the entire lot shall be rejected. If one or more minor defects are found in any sample, the contractor has the option of screening 100 percent of the rejected lot for the defective characteristic(s), or providing a new lot, which shall be inspected in accordance with the sampling plan contained herein.

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TABLE V. Sampling for Quality conformance inspection/verificat ion.

Lot Size	Major Characteristic	Minor Characteristic
2 to 50	All	3
51 to 90	All	6
91 to 150	125	7
151 to 280	125	10
281 to 500	125	11
501 to 1,200	125	15
1,201 to 3,200	125	18
3,201 to 10,000	192	22
10.001 to 150,000	294	29

4.5 Quality conformance verification inspection. Upon completion of the tests and examination specified in 4.4 (see 6.4) , quality conformance verification, when specified (see 6.2), shall be performed by a laboratory specified by the government.

4.5.1 QualitY conformance verification samples. The quality conformance verification inspection shall consist of the tests and examinations specified in Table IV. The sample size shall be in accordance with Table V. Each preserver selected as a sample unit shall be identified by its assigned serial number (see 3.8.2) Additionally , the serial numbers of the units within the lot represented by the sample units shall be furnished to the inspection facility.

4.5.2 Quality conformance verification approval . Final acceptance of the lot from which the samples were selected shall be based upon successful completion of the inspection program by the cognizant Government quality assurance representative specialist. Upon completion of this inspection program the samples will be returned to the contractor.

4.6 Inspection conditions

4.6.1 Atmospheric conditions. Unless otherwise specified, all the inspections required by this document shall be conducted at an atmospheric pressure of 28 to 32 inches (711.2 to 812.8 millimeters (mm)) of mercury and at a temperature of 77 ± 18 degrees Fahrenheit (“F) (25 ± 10 degrees Celsius (“C)). If the final values of the ambient temperature or barometric pressure at the end of the four hour air pressure inspection are different from the initial values recorded at the start of the inspection, the following corrections shall be made to the final pressure readings in psi gauge.

4.6.1.1 Temperature correction. For each degree Fahrenheit rise in temperature , 0.031 psi gauge shall be subtracted from the final pressure reading. For each *degree* Fahrenheit drop in temperature, 0.031 psi gauge shall be added to the final pressure reading. The corresponding correction per degree Celsius is 0.056 psi gauge.

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4,6.1.2 Barometric Pressure correction. For each 0.1 inch of mercury rise in barometric pressure, 0.049 psi gauge shall be added to the final temperature-corrected pressure reading. For each 0.1 inch of mercury drop in barometric pressure, 0.049 psi gauge shall be subtracted from the final temperature-corrected pressure reading.

4.6.2 Pressure measurement. The pressure shall be measured by means of a mercury manometer or low pressure gauge calibrated in tenths psi gauge.

4.6.3 Inspection area and equipment. The area in which the life preservers are inspected, shall be adequately protected co preclude damage to the units. The area and inspection equipment shall be free of sharp or rough edges, burrs, pro-trusions, and so forth, which will cut, tear, or damage the life preservers or components .

4.6.4 Air. When use of air is specified in an inspection, the air used shall not contain any oil or condensed water vapor and shall be breathable quality.

4.7 Inspection methods.

4.7.1 Visual and Packaging examination.

4.7.1.1 Visual. Every preserver shall be visually examined for defects to determine conformance to the requirements of 3.10.6 using the examination contained in Table VI.

TABLE VI. Classification of defects for visual examination of the life preserver.

DEFECT	CLASSIFICATION	
	MAJOR	MINOR
1. Any non-specified hole, cut, patch, or burn.	x	
2. Any fabric damaged, bruised, abraded, containing imperfections , or is otherwise defective. 1/	x	
3. Any stitching in the inflatable section of the life preserver.	**x**	
4. Manifold stem assembly not in accordance with requirements ; not located as specified, or inoperable.	**x**	
5. Any component, component part, or required operation omitted; or *any* operation improperly performed not herein classified. 2/	x	
6. Any component not as specified, or any defect of a component or assembly not herein classified. 2/	x	

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TABLE VI. Classification of defects for visual examination of the life preserver (cont'd).

	DEFECT	CLASSIFICATION	
		MAJOR	MINOR
7.	Any seam tape not applied to seams, as specified. 2/	x	
8.	Stitching and sewed seam construction not as specified.	x	
9.	Oral inflation assembly dislocated, bent, distorted, or not operable. Oral tube clamp not installed securely.	x	
10.	Any cut edge of the uncoated nylon fibrous materials not seared or containing sharp edges.	**x**	
11.	Oral inflation assembly or components not specified type; oral valve not locked in the closed position.	x	
12.	Any open seam in the inflatable bladder.	X	
13.	Any metal component improperly finished or containing nicks, burrs, dents, sharp edges or rough surfaces as specified.	x	
14.	Any inflatable section seam construction which does not meet the minimum requirements specified.	**x**	
15.	Any hook and pile fastener missing or incorrectly located.		x
16.	Any channels or voids in any cementing operation. 1/		x
17.	Any reflective tape missing or incorrectly located.		x
18.	One or more spots or stains. 3/		x
19.	Color of any component not as specified.		x
20.	Any required markings missing, illegible, incomplete, incorrect or improperly attached or located.		**x**
21.	Any inspection record patch or warning tag missing, illegible, incomplete, incorrect, or improperly attached or located.		x
22.	Excessive wrinkling caused by any cementing or heat sealing.		x
23.	Any inflatable section seam construction which exceeds the maximum requirements specified.		x
24.	Any seam not butted, as specified.		x
25.	Cement on the cloth surfaces around the patches below a minimum length of 1/8 inch or in excess of 3/8 inch.		

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TABLE VI. Classification of defects for visual examination of the life preserver (cent'd) .

DEFECT	CLASSIFICATION	
	MAJOR	MINOR
26. Any clot or mass of adhesive.		x
27. Overlap of any seam tape in excess of requirements.		x
28. Any seam tape, patch, or attachment separating.		x
29. Life preserver not properly folded.		x

1/ The defect shall be classified major if it is in the inflatable section; otherwise , it is to be classified minor.
2/ The defect shall be classified major when it seriously affects service ability or function; otherwise, it is to be classified minor.
3/ The defect shall be classified as minor when it does not seriously affect serviceability or function; otherwise, it is to be classified major.

4.7.1.2 Packaging. Each of the fully prepared shipping containers, containing life preservers, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects as specified in Table VII, shall be used to enumerate the defects found.

TABLE VII. List of defects for packaging packing and marking.

Item	Defect
Exterior and interior markings	Missing, incorrect , incomplete, illegible, of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers.
Packaging and packing material	Any non-conforming component; any component damaged or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure **of** the unit package, container flaps, loose strappings, etc.-; bulging or distortion of the containers; unit container contains metal fastenings or stitches.
Exterior and interior weight	Number per container is more or less than required; gross or net weight exceeds the requirements ; item not folded as specified.

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4.7.2 Tests. The complete preserver shall be tested in accordance with this specification.

4.7.2.1 Operation with inflator assembly and CO_2 cylinder. The operation of the preserver shall be tested by releasing the CO_2 cylinder gas into the packed (bladder folded into the cover) life preserver manually using a Government Furnished inflation assembly. The CO_2 cylinder shall conform to MIL-C-25369, Type II. The preserver shall be observed for conformance to the requirements of 3.10.2. If the buoyancy inspection is not required, the preserver shall be completely deflated by unlocking the oral inflation valve, depressing the mouthpiece and discharging the carbon dioxide. During the discharge of carbon dioxide, the mouthpiece shall be released a minimum of three times and observed for conformance to the requirements of 3.10.2.1.

4.7.2.2 Adhesion. The adhesion test shall be in accordance with 4.5.1 of SAE AMS 3272. The test results shall meet the requirements of 3.10.1. Specimens similar to the seams of the bladder shall be used for this test, not fully constructed bladders.

4.7.2.3 Buoyancy. The buoyancy inspection shall be performed in conjunction with the operation inspection (see 4.7.2.1) while the preserver is still inflated with carbon dioxide. Remove the inflation assembly and CO_2 cylinder from the life preserver. The preserver, with a 32 pound weight (solid steel or lead) attached, shall be placed in clean, fresh water having a temperature of 75 ± 5°F (23 ± 3°C), and observed for conformance to the requirement of 3.10.3. Upon completion of the test and inspection, the preserver shall be completely deflated through its oral inflation assembly in the manner outlined in 4.7.2.1 and observed for conformance to the requirements of 3.10.2.1.

4.7.2.4 Over Pressure. The carbon dioxide cylinder shall be removed for this inspection. The preserver shall be inflated with air through its oral inflation assembly to a pressure of 5 psi gauge. The air shall then be securely shut off and after a minimum of 15 minutes the pressure shall be checked and readjusted, if necessary, to the original pressure of 5 psi gauge. Fifteen minutes after the check and readjustment, the pressure in the preserver shall not be less than 4.5 psi gauge. Upon completion of this test and inspection, the preserver shall be completely deflated in preparation for the leakage inspection (see 4.7.3.5).

4.7.2.5 Leakage. The bladder, without the inflation assembly and the CO_2 cylinder installed, shall be inflated with air through its oral inflation assembly to a pressure of 2 psi gauge. The air shall then be securely shut off and after a minimum of 15 minutes the pressure shall be checked and readjusted, if necessary, to the original pressure of 2 psi gauge. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 4 hours after the check and readjustment, the pressure shall be measured and corrected for any change in the temperature and barometric pressure (see 4.6.1.1 and 4.6.1.2). The preserver shall conform to the requirements of 3.10.5. Upon completion of the inspection, the life preserver shall be completely deflated.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.

5.1 General.

5.1.1 Navy fire-retardant requirements.

a. Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category I - general use.

Level C - Type I - non-weather-resistant .
Category I - general use.

b. Fiberboard. Fiberboard used in the construction of interior (unit and intermediate) and exterior fiberboard boxes including interior packaging forms shall conform to the class-domestic/fire retardant or class-weather resistant/fire retardant materials requirement as specified (see 6.2) of PPP-F-320.

5.2 Preservation. Preservation shall be level A, C or commercial, as specified (see 6.2).

5.2.1 Level A. Each life preserver, properly folded as specified in 3.11, shall be completely wrapped in a minimum 30 pound basic weight fire retardant kraft paper conforming to UU-P-268, type II, grade C or A or A-A-1894 at the suppliers option. The wrap shall be secured with pressure sensitive water resistant tape. Each wrapped life preserver shall be unit packed into a water resistant, folding, set-up, or metal edged paperboard or fire retardant fiberboard unit container meeting the unit and intermediate container requirements of MIL-STD-2073-1. Fiberboard box closure shall be in accordance with method V of the appendix to the box specification. Container selection shall be at the option of the contractor. Each life preserver unit pack shall meet the requirement for method 111 of MIL-P-116.

5.2.2 Level C. Each life preserver shall be unit protected as specified for level A except that the unit containers shall be as follows:

a. The paperboard containers shall be of the domestic or non-weather resistant type, class, or variety as applicable and,

b. The fiberboard containers shall be of the class-domestic/fire-retardant material (see 5.1.1 (b)). The box closure shall be in accordance with method I using pressure sensitive, adhesive tape.

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5.2.3 Commercial. Commercial packaging (cleaning, preservation, cushioning, unit and intermediate package) of life preservers folded as specified in 3.11 shall be in accordance with-ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C or Commercial, as specified (see 6.2).

5.3.1 General requirements for levels A. B and C. Containers selected (see 5.3.2), shall be of minimum weight and cube consistent with the protection required, of uniform size, and contain identical quantities of identical preservers.

5.3.2 Levels A. B and C containers. Life preservers, preserved as specified (see 5.2), shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with the exterior shipping container requirements of MIL-STD-2073-1 and herein. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Caseliners, closure and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers containing life preservers preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that the class weather-resistant including fire retardant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and class domestic/fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Commercial. Life preservers preserved as specified (see 5.2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall have a minimum of two, 3-inch by 4-inch nominal wood skids laid flat, or a skid or still type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage.

5.4 Marking, levels A. B. C and commercial. - In addition to any special marking required (see 6.2) , interior packs and shipping containers shall be marked including bar coding for shipment, date of manufacture (month and year), stowage and storage in accordance with MIL-STD-2073-1.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The automatically inflatable utility life preserver is to be worn by personnel working over the side; performing duties during underway replenishment ; serving as boat crews and manning battle stations.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification, including any amendments .
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
(c) Serial numbers to be assigned by the contractor (see 3.8.2).
(d) Government part number, national stock number, and quantity.
(e) When first article inspection is required (see 3.1)
(f) Name and address of the first article inspection laboratory (see 4.3).
(g) Quality conformance verification samples (see 4.5.1)
(h) Name and address of the quality conformance verification inspection laboratory (see 4.5.1); and the name and address of the Government activity responsible for conducting the quality conformance verification inspection program (see 4.5.2) .
(i) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(j) Requirements for MIL-STD-129 marking or for special marking (see 5.3).
(k) Certification of manufacture of materials.

6.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.5.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data,except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form *1423.*

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DID-RELI-80303	Quality Conformance Inspection & Test	----
4.5	DI-NDTI-80604	Test Report	----

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The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), q ust be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, q ust furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Subject term (key word) listing.

Bladder
Oral inflation valve
Vest
Hood
Face shield

Preparing activity:
Navy - SH
(Project 4220-N360)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-24778(SH)	17 DECEMBER 1992

3. DOCUMENT TITLE
LIFE PRESERVER, UTILITY, AUTO-INFLATABLE, MK 5, MOD 0

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
TECHNICAL POINT OF CONTACT (TPOC): A. ULIANA, SEA 05G1, NAVAL SEA SYSTEMS COMMAND	(1) Commercial (703) 602-7217	(2) AUTOVON 8-332-7217
c. ADDRESS (Include Zip Code) ATTN: SEA 05Q42 2541 NATIONAL CENTER BLDG. 3 WASHINGTON, DC 20362-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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